INCH-POUND

CCC-C-442E
November 30, 1989
SUPERSEDING
CCC-C-442D
April 5, 1974

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON, BLEACHED

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bleached cotton duck utilized in the manufacture of clothing items (see 6.1).

1.2 Classification. The bleached cotton duck shall be of the following types as specified (see 6.1 and 6.2).

Type I	-	5.0 ounces per square yard
Type II	-	6.5 ounces per square yard
Type III	-	8.2 ounces per square yard
Type V	-	7.0 ounces per square yard

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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Federal Specifications:

V-T-276	-	Thread, Cotton
PPP-P-1135	-	Packaging and Packing of Duck Fabrics (Cotton: Synthetic Fiber: Cotton - Synthetic Fiber Blends)

Federal Standards:

Fed. Std. No. 4	-	Glossary of Fabric Imperfections
Fed. Std. No. 191	-	Textile Test Methods

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Federal Regulations:

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, shall supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Standard Sample. The bleached and finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Yarn. The yarn shall be made from cotton which has been carded, drawn, and twisted into single or plied yarns as specified in table I.

3.4 Color. The cloth shall be chemically bleached white and may be supplemented with optical brighteners which fluoresce to the blue region. The degree of whiteness shall match the standard sample.

3.4.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.4.2 Colorfastness. The finished bleached cloth shall not discolor to a degree greater than that shown by the standard sample, or the discoloration shall be no lower than the rating of "Fair" when tested as specified in 4.4.3.

3.5 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.3.

TABLE I. Physical requirements

Type	Weight per square yard minimum	Yarn ply		Yarns per inch minimum		Breaking strength minimum	
		Warp	Filling	Warp	Filling	Warp	Filling
	Ounces					Pounds	Pounds
I	5.0	2	2	77	44	130	75
II	6.5	1	2	88	24	110	60
III	8.2	2	2	58	48	135	100
V	7.0	2	2	52	34	80	48

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3.5.1 Width. The width of the cloth shall be as specified (see 6.2). The minimum acceptable width shall be inclusive of the woven selvage when fly shuttle looms are used or when tuck-in selvage is used on shuttleless loom and exclusive of the lock-in fringed selvage when a shuttleless leno lock-in fringe selvage is used. Cut raw edges shall not be permitted.

3.6 Weave. The weave for types I, III and V cloths shall be plain, one up, one down. The weave for type II cloth shall be an oxford weave with two ends weaving as one and alternating in plain-weave order with each pick. The use of the fly shuttle or a shuttleless loom is permitted.

3.7 Finish. The cloth shall be desized, singed, and scoured, bleached and mercerized.

3.7.1 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material for the finished cloth shall not exceed 2.0 percent when tested as specified in 4.4.3.

3.8 Shrinkage. The cloth shall be preshrunk so that the residual shrinkage shall be not more than 2.0 percent in either the warp or filling direction when tested as specified in 4.4.3. The preshrinking process used shall not be identified in any manner on the cloth, the ticket, or on the package.

3.9 pH. The pH of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.3.

3.10 Seam efficiency. The seam efficiency of the finished cloth shall be not less than 80 percent when tested as specified in 4.4.3.

3.11 Length and put-up. Unless otherwise specified (see 6.2) the cloth shall be in continuous lengths, each not less than 50 yards. The cloth shall be put-up on rolls as specified in 5.1.

3.12 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.13 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point values.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 through 4.4.2.3 and tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the cloth conforms to the singing requirement specified in 3.7.

4.4.2 End item examination.

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4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. All defects, as defined in section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.4.2.1.1 except that only those slubs and knots which exceed the limits shown in figure 1 of FED-STD-4 shall be scored. No linear yard (increments of 1 yard on the measuring device or the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 35 points. The lot shall be unacceptable if the points per yard 100 square yards of two or more individual rolls exceeds 53 points. If one roll exceeds 53 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 53 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Width less than specified
Overall uncleanness
Objectionable odor

4.4.2.2 Length examination.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

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4.4.2.3 Examination for non-identification of preshrinking process, face identification and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample.

Face identification missing from either or both ends
Face identification on wrong side
Preshrinkage process identified by name or trademark on cloth or ticket
Not labeled, ticketed, or invoiced in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act

4.4.3 End item testing. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width of the finished cloth. The sample size shall be as listed in table II. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final results.

TABLE II. Sample size

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Colorfastness	3.4.2	5660 1/ 2/
Weight	3.5	5041
Yarn ply:		
Warp	3.5	Visual 2/
Filling	3.5	Visual 2/
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050

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TABLE III. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Weave	3.6	Visual 2/
Singed	3.7	3/
Mercerization	3.7	Microscopic examination for swelling of cotton fiber 2/
Nonfibrous material	3.7.1	2611
Shrinkage in laundering	3.8	5550
pH	3.9	2811
Seam efficiency	3.10	5110 4/

1/ Method 5660 shall be used, except that the exposure of the specimen and standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass." Any specimen showing discoloration greater than that of the standard sample shall be rated "fail." Whether a standard sample is or is not available, the discoloration shall be considered satisfactory if it is not worse than the rating of "Fair" in the scale indicated in Method 5660.

2/ One determination for each sample unit and the result reported as "pass" or "fail."

3/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the requirements stated.

4/ The thread shall be cotton conforming to type IA2 of specification V-T-276. The needle thread shall be ticket No. 24, 3 ply and the looper thread shall be ticket No. 60, 3 ply. The needle size shall measure 0.049 ± 0.001 inch across the blade at the eye.

4.4.4 Packaging inspection. The inspection shall be made in accordance with the quality assurance provisions of PPP-P-1135.

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5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial preservation. The cloth shall be put-up on rolls and preserved in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial packing. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1135.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Types I, II, and III bleached cotton ducks are used in white uniforms for personnel of all Government departments. Type V bleached cotton duck is utilized as the base cloth for Smock, Mens's, Dental, Operating; and Smock, Men's, Medical Assistant.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Width required, if other than specified (see 3.5.1).
f. Length required, if other than specified (see 3.11).
g. Levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Clothing
Textile fabrics
Uniforms

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - NU

Review activities:

Army - MD
DLA - CT

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS

Preparing Activity:

Army - GL

(Project 8305-0308)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
CCC-C-442F	CLOTH, DUCK, COTTON, BLEACHED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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